

Galois theory Group Project

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1 Ruler and Compass Constructions

Recall the tower law: let $K \subseteq L \subseteq M$ be fields, then $[M : K] = [M : L][L : K]$. This could play an important role answering some questions in geometry which stymied the ancients. In particular, we want to show that certain things could not be constructed using a compass and a ruler. We will go through three famous examples:

- (i) Doubling the cube: Given a cube, constructing a cube whose volume is double the volume of the original cube.
- (ii) Trisecting an angle: Given an angle, dividing the angle into three identical angles.
- (iii) Squaring the circle: Given a circle, constructing a square with the same area i.e. construct length π given 1.

In addition, we will investigate the ruler and compass construction of a regular n -gon. For example, 17-gon by Gauss!

Definition 1.1 (Constructible points). Let $S \subseteq \mathbb{R}^2$ be finite. We say $T \in \mathbb{R}^2$ is **one-step constructible** from S , if T is a point of intersection of two curves C_1 and C_2 , where C_i is either

1. a line drawn between two distinct points $s_1, s_2 \in S$.
2. a circle, with centre $s \in S$, and radius being the distance between two points $s_1, s_2 \in S$.

We say $R \in \mathbb{R}^2$ is **constructible** from S , if there is a sequence $R_1, R_2, \dots, R_n = R$ such that each $R_i \in \mathbb{R}^2$ is one-step constructible from $S \cup \{R_1, \dots, R_{i-1}\}$.

Example. Let $S = \{(0,0), (1,0)\}$. It's easy to see that $(n,0)$ for all $n \in \mathbb{Z}$ are constructible from S . In fact, one can show that all the points of the form (n,m) for $n, m \in \mathbb{Z}$ are constructible from S .

Remark. Note that we can only construct countably many points from $\{(0, 0), (0, 1)\}$, since we always have a finite number of options from a finite number of points, so there are certainly points we cannot construct.

Definition 1.2. Let $S \subseteq \mathbb{R}^2$ be finite. Define the field of S by

$$\mathbb{Q}(S) = \mathbb{Q}(\{\text{coordinates of points in } S\}) \subseteq \mathbb{R}$$

i.e. if $S = \{p_1, \dots, p_n\}$ where $p_i = (x_i, y_i) \in \mathbb{R}^2$ then

$$\mathbb{Q}(S) = \mathbb{Q}(x_1, y_1, x_2, y_2, \dots, x_n, y_n) \subseteq \mathbb{R}.$$

The key theorem we will use to prove our results is:

Theorem 1.3. Let $S \subseteq \mathbb{R}^2$ be finite. Then

- (i) If R is constructible in one step from S , then $[\mathbb{Q}(S \cup \{R\}) : \mathbb{Q}(S)] = 1$ or 2.
- (ii) Let $T \subseteq \mathbb{R}^2$ be finite such that every point in T is constructible from S . Then $[\mathbb{Q}(S \cup T) : \mathbb{Q}(S)] = 2^k$ for some k .

Proof.

- (i) Suppose R is constructible in one step from S , then there are distinct lines or circles C_1, C_2 constructed from S using ruler and compass where $R \in C_1 \cap C_2$. Let $R = (r_1, r_2)$ then, there are three situations:

- (1) If C_1, C_2 are two straight lines then they are given by the equations

$$\begin{aligned} C_1 : y &= a_1x + b_1 \\ C_2 : y &= a_2x + b_2 \end{aligned}$$

where $a_1, a_2, b_1, b_2 \in \mathbb{Q}(S)$. If $a_1 = a_2$ and $b_1 \neq b_2$ then C_1, C_2 are parallel so this does not make any sense since they do not intersect. If $a_1 \neq a_2$ then

$$r_1 = \frac{b_1 - b_2}{a_2 - a_1} \in \mathbb{Q}(S), \quad r_2 = a_1 r_1 + b_1 \in \mathbb{Q}(S)$$

so $r_1, r_2 \in \mathbb{Q}(S)$, therefore we have $[\mathbb{Q}(S \cup \{R\}) : \mathbb{Q}(S)] = 1$.

- (2) Suppose without loss of generality that C_1 is a circle and C_2 is a straight line. Then they are given by the equations:

$$\begin{aligned} C_1 : (x - a)^2 + (y - b)^2 &= c^2 \\ C_2 : y &= dx + e \end{aligned}$$

where $a, b, c^2, d, e \in \mathbb{Q}(S)$. Then

$$(r_1 - a)^2 + (dr_1 + e - b)^2 = c^2$$

$$p(r_1) := (1 + d^2)r_1^2 + (2d(e - b) - 2a)r_1 + (a^2 + (e - b)^2 - c^2) = 0.$$

Let $p'(t) := \frac{p(t)}{1 + d^2}$ be the monic polynomial in $\mathbb{Q}(S)[t]$.

(a) If $p'(t)$ is irreducible, then $[\mathbb{Q}(S)(r_1) : \mathbb{Q}(S)] = 2$ as $p(t)$ is the minimal polynomial of r_1 over $\mathbb{Q}(S)$. Since $r_2 = dr_1 + e \in \mathbb{Q}(S)(r_1)$, it follows that $[\mathbb{Q}(S \cup \{R\}) : \mathbb{Q}(S)] = [\mathbb{Q}(S)(r_1, r_2) : \mathbb{Q}(S)] = [\mathbb{Q}(S)(r_1) : \mathbb{Q}(S)] = 2$.

(b) If $p'(t)$ is reducible, it can be factorized into two linear factors, so the root $r_1 \in \mathbb{Q}(S)$ which implies $[\mathbb{Q}(S \cup \{R\}) : \mathbb{Q}(S)] = [\mathbb{Q}(S)(r_1, r_2) : \mathbb{Q}(S)] = [\mathbb{Q}(S) : \mathbb{Q}(S)] = 1$.

(3) Finally, suppose C_1, C_2 are two distinct circles with equations:

$$C_1 : (x - a)^2 + (y - b)^2 = c^2 \quad (\text{A})$$

$$C_2 : (x - d)^2 + (y - e)^2 = f^2 \quad (\text{B})$$

Note that by considering the equivalent system of (A) and (A)-(B), we return back to case (2) which we have discussed previously.

(ii) This is an immediate consequence by part(i) and induction, by using the tower law. $\square$

Corollary 1.4. It is impossible to "double the cube".

Proof. Consider the cube with unit side length, i.e. we are given the set $S = \{(0, 0), (1, 0)\}$. Then doubling the cube would correspond to constructing a side of length $\sqrt[3]{2}$. Thus, we need to construct a point $R = (\sqrt[3]{2}, 0)$ from S . By Theorem 1.3, we need

$$[\mathbb{Q}(S \cup \{R\}) : \mathbb{Q}(S)] = 2^k \text{ for some } k.$$

But note that $\mathbb{Q}(S) = \mathbb{Q}$ so $\mathbb{Q}(S \cup \{R\}) = \mathbb{Q}(\sqrt[3]{2})$. But we know

$$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3.$$

which is obviously not a power of 2, a contradiction. $\square$

Corollary 1.5. It is impossible to 'trisect an angle'.

Proof. From $S = \{(0, 0), (1, 0)\}$, we could easily construct the angle 60° . If this can be trisected, then it would be possible to construct the point $(x, y) = (\cos 20^\circ, \sin 20^\circ)$. Recall that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$. If $x = \cos 20^\circ$ then $4x^3 - 3x = \cos(60^\circ) = \frac{1}{2}$ i.e. $8x^3 - 6x - 1 = 0$. Write $x = t - 1$:

$$8(t - 1)^3 - 6(t - 1) - 1 = 8t^3 - 24t^2 + 18t - 3$$

which is irreducible by Eisenstein with prime 3. Thus, $[\mathbb{Q}(x), \mathbb{Q}] = 3$, a contradiction again. $\square$

Corollary 1.6. It is impossible to 'square the circle'.

Proof. This means π is constructible from $S = \{(0, 0), (1, 0)\}$. Since π is transcendental (Proof omitted), $[\mathbb{Q}(\pi) : \mathbb{Q}] = \infty$ which is not a power of 2. $\square$

2 Constructible n-gon

Now we continue with the general n-gon.

With a similar approach of Corollary 1.5, a regular n-gon is constructible if we can construct the point $(\cos(\frac{2\pi}{n}), \sin(\frac{2\pi}{n}))$ from $S = \{(0, 0), (0, 1)\}$. In the complex plane, it is just $\zeta_n = \cos(\frac{2\pi}{n}) + i \sin(\frac{2\pi}{n})$ which is also the nth primitive root.

By Theorem 1.3, we know that if this point is constructible from $S = \{(0, 0), (0, 1)\}$, then $[\mathbb{Q}(S \cup \zeta_n) : \mathbb{Q}] = [\mathbb{Q}(\zeta_n) : \mathbb{Q}]$ is some power of 2. On the other hand, the degree is also $\varphi(n)$, where $\varphi(n)$ is the Euler totient.

Now we have the following fact:

Proposition 2.1. The regular n-gon is constructible, then the odd prime factors of n are Fermat primes, i.e $F_r = 2^{2^r} + 1$.

Proof. If $n = 2^k p_1^{a_1} \dots p_m^{a_m}$ with odd primes p_i , then

$$\varphi(n) = 2^{k-1} (p_1 - 1) p_1^{a_1-1} \dots (p_m - 1) p_m^{a_m-1}$$

If it is some power of 2. Then all $a_i = 1$ and each $p_i - 1$ is some power of 2. $\square$

Also, by using Galois groups (Proof omitted), we find the other direction is true:

Theorem 2.2 (Gauss–Wantzel Theorem). A regular n-gon is constructible iff the above condition is true.

Proof. Omitted. $\square$

Therefore in conclusion, a n-gon is constructible iff $\frac{\varphi(n)}{2}$ is some power of 2.